Generated sets

MatterGen
Base
Fine–tuned

Con-CDVAE
Base
Fine–tuned

CrystalFormer
Base
Fine–tuned

CrystaLLM
Base
Fine–tuned

Crys-JEPA
Base
Refined

Training data
(for magnetic domain fine–tuning)
• MAGNDATA
• NovoMag
• NEMAD

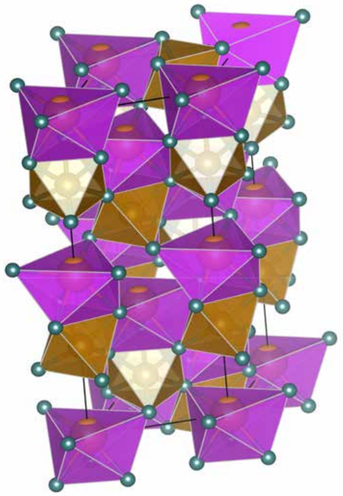
Generated–data workflow

Retrained structures for downstream analysis

CIF harmonization

Structural quality control

Genetayed structures / native outputs



Example of generated CIF crystal

Reference data
(for reference space screening)
Materials Project reference structures (AFM, FM, FiM, NM)

Reference space screening
Descriptor construction
kNN distances
Contrast scoring magnetic-reference contrasts

Property prediction
CHGNet energy
predicted energy per atom
DXMag magnetic moment
composition–based predicted moment

Magnetic descriptors
Composition 3d, selected 4d/5d, lanthanides
Local magnetic geometry coordination and neighbor distances
Symmetry magnetic site and crystallographic symmetry

Candidate–level DFT validation
Selected candidates
DFT calculations
DFT validation

Integrated population level analyses
Structural retention and rejection profiles
Magnetic domain composition and local environments
Reference space enrichment
Descriptor space PCA
Energy–magnetism landscape
Chemical composition and diversity